Cross-Modal Robust EEG Biomarkers for Sustained Attention in Multimodal Learning

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***Abstract*—Real-world attention monitoring requires EEG biomarkers that generalize across varying sensory contexts; however, standard biomarkers often overfit to the specific modality used during training. A multimodal learning dataset (Text, Audio, Video) was collected using a consumer-grade Emotiv EPOC X headset to study modality-robust attention decoding. This paper proposes a *novel* feature-ranking mechanism that balances the discriminative power of candidate biomarkers against inter-modality distributional divergence, quantified using the 1-Wasserstein metric. The resulting criterion isolates features that maximize attention-state separability while minimizing domain shifts induced by changing sensory inputs. Under a fixed-biomarker Leave-One-Subject-Plus-Leave-One-Modality-Out (LOSO–LOMO) evaluation, an SVM classifier achieved a *balanced accuracy* of 0.64 using an optimal 26-feature set, outperforming a redundancy-pruned baseline feature set (0.59). In a leakage-controlled Leave-One-Subject-Out (LOSO) setting with fold-local fitting and selection, the median *balanced accuracy* reached 0.66 across folds.**

***Index Terms*—EEG; sustained attention; modality invariance; consumer EEG; Wasserstein distance; LOSO; LOMO; feature stability; redundancy pruning**

I. Introduction

A. Background and Motivation

Attention is foundational to effective learning, yet many EEG-based attention detectors are developed and validated in a single stimulus modality. Models that perform well during reading can degrade when users switch to listening or watching because EEG feature distributions shift with sensory input [1], [2]. This limits passive brain–computer interfaces (BCIs) in realistic settings where users alternate between text, audio, and video. We therefore hypothesize that a small "common core" of EEG features can indicate sustained attention across modalities [3]; the goal is to identify such modality-stable biomarkers to enable attention monitoring without modality-specific re-training.

This hypothesis is neurophysiologically plausible: sustained attention reflects top-down executive control supported by the frontoparietal attention network, which is partly decoupled from modality-specific visual and auditory processing. Therefore, although reading, listening, and watching engage different sensory pathways, sustained executive engagement may impose comparatively invariant, large-scale signatures in scalp EEG.

Validating cross-modal biomarkers is challenging with consumer-grade EEG and limited cohorts. Wearable systems (e.g., Emotiv EPOC X) are more susceptible to noise and artifacts than research-grade recordings [4], [5], and small-N studies increase the risk of overfitting when using deep models. In addition, epochs within a subject are correlated, so treating each epoch as an independent sample can inflate significance. Accordingly, this paper emphasizes subject-level inference and strict fold-local fitting/selection to ensure that results reflect generalization to unseen participants and modalities [6].

B. Related Work

Prior work decodes sustained attention from EEG in controlled paradigms (e.g., continuous speech tracking) using spectral features and temporal response functions [7], [8]. Many pipelines exploit stimulus-locked structure (e.g., envelope reconstruction), which is modality-specific and does not directly transfer to self-paced tasks such as reading [24]. Feature-based attention recognition using time-, frequency-, and nonlinear-dynamics descriptors has also been reported [44]–[47]. While transfer learning has been extensively used to target cross-subject variability in EEG [6], [9], robustness toward stimulus-modality shifts (i.e., visual vs. auditory) has received significantly less attention [1]. As a result, standard feature-selection methods rarely penalize modality-driven distribution changes, leading to poor generalization in multi-modal environments [21].

This work addresses this gap by using the Wasserstein distance to quantify how feature distributions vary across learning modalities. Instead of explicitly aligning domains via complex transformations, the proposed *novel* method uses this distance to regularize feature ranking, prioritizing biomarkers that distinguish attention from relaxation while remaining stable across modalities. This approach draws inspiration from Optimal Transport frameworks used in domain adaptation, where minimizing the Wasserstein distance encourages feature invariance across disparate distributions [10], [11].

C. Contributions

The contributions of this work are as follows:

1) A multimodal stimuli EEG dataset recorded with Emotiv EPOC X (14 channels; $256\,\mathrm{Hz}$) with fixed, non-overlapping $6\,\mathrm{s}$ epochs for text, audio, and video learning plus a relaxed baseline;
2) A biomarker discovery pipeline that combines redundancy pruning and *novel* Wasserstein-based modality-stability ranking, with subject-level inference and FDR control; and

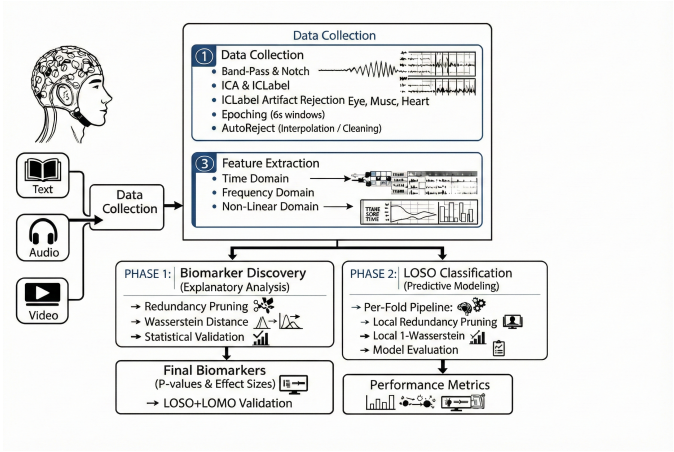


Fig. 1. Analysis pipeline: preprocessing, feature extraction and redundancy pruning, *novel* Wasserstein-based feature ranking, and LOSO/LOMO evaluation.

3) A cross-modal evaluation protocol that includes strict, leakage-controlled leave-one-subject-out (LOSO) classification and a leave-one-subject-plus-leave-one-modality-out (LOSO–LOMO) stress test to assess robustness to unseen modalities.

II. Methodology

A. EEG data acquisition

Fourteen healthy adults (mean age: 20 years) participated with written informed consent. EEG was recorded with an Emotiv EPOC X (14 channels; 256 Hz; AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4) using EmotivPRO software [28] during attentive learning tasks (text: silent reading; audio: listening; video: watching a lecture) and a relaxed baseline. The baseline consisted of three $\sim$1 min trials per participant (total $\sim$3 min). Task durations were $\sim$12 min (audio/video) and 8–12 min for text (depending on reading speed); learning materials were selected for comparable difficulty across modalities.

Labels were assigned from segment type and instructions (*attention* during learning tasks vs. *relaxed* during baseline). The paradigm targets sustained attention (not event-related processing); accordingly, reliable ERP components are not expected. A post-task 10-item quiz and self-report were collected but not used as epoch-level labels. Fixed, non-overlapping 6 s epochs were extracted with median epoch counts per subject as $\sim$55 (for text attention), $\sim$70 (for audio/video attention), and $\sim$23 (for relaxed baseline); also class imbalance is handled during evaluation (Section II-F).

B. Preprocessing and artifact handling

Preprocessing was implemented using *MNE-Python* [29]. In brief, artifacts were handled by first excluding severely noisy segments, then fitting ICA, and finally removing artifactual ICs before epoch-level repair/rejection as mentioned below:

TABLE I
Preprocessing parameters (key settings).

Stage	Configuration	Specification
Bandpass filter	Range	1 – 45 Hz
Notch filter	Frequency	50 / 60 Hz
Pre-ICA exclusion	Peak-to-peak threshold	500 μV
ICA	Algorithm	Extended Infomax
	PCA (pre-ICA)	95% variance retained
Artifact labeling	ICLabel threshold	0.8 (probability)
Epoching	Duration	6 s (non-overlapping)
Artifact rejection	Method	AutoReject
Re-referencing	Reference	Common average

1) **Filtering:** Zero-phase FIR bandpass (1–45 Hz) and notch filtering at 50/60 Hz.
2) **Bad-channel screening:** Flatline detection.
3) **Pre-ICA exclusion:** Remove segments (within an epoch) with peak-to-peak amplitude $> 500\,\mu$V.
4) **ICA:** Extended Infomax ICA, with PCA retaining 95% variance [30].
5) **IC removal:** ICLabel-based removal of ocular, myogenic, cardiac, line-noise, and channel-noise components (probability ≥ 0.8) [31], [32].
6) **Interpolation:** Interpolated bad channels *after* IC removal.
7) **Epoch repair/rejection:** AutoReject (data-driven, cross-validated amplitude-threshold selection with optional interpolation of bad sensors in epochs) [33], [43].
8) **Re-referencing:** Common-average reference (after interpolation and epoch repair/rejection).

Key parameter values (held constant across subjects) are listed in Table I.

Non-overlapping 6 s epochs for both attention and relaxed conditions were extracted. This duration was selected to support reliable band-power and complexity estimates, ensuring that slow fluctuations in sustained attention are effectively captured.

C. Feature extraction

A diverse set of per-epoch features was extracted to capture complementary EEG characteristics. Unless specified as a spatial aggregate, each feature was computed per channel (14 channels total). Numbers in parentheses denote the total feature count across all channels for each feature family.

1) **Time-domain (112):** Root mean square (RMS), variance, skewness, kurtosis, line length, zero-crossing rate (ZCR), and Hjorth activity, mobility, and complexity [13].
2) **Frequency-domain (350):** Absolute and relative power in standard bands (δ through γ), band-power ratios (e.g., θ/β ratio), spectral entropy, spectral slope, spectral edge frequency (SEF), and spectral centroid, estimated via Welch's method [14].
3) **Complexity/nonlinear (84):** Higuchi, Katz, and Petrosian fractal dimensions, permutation entropy, sample entropy, and Lempel–Ziv complexity [15]–[17].

4) **Energy (56):** Teager–Kaiser energy operator (TKEO) applied to broadband and filtered θ, α, β bands [23].
5) **Spatial aggregates (9):** Regional power averages, hemispheric asymmetry (Left vs. Right), and frontal–parietal alpha ratios.

D. PHASE 1: Biomarker discovery

1) Subject-wise normalization and leakage control: Features were z-scored within subject to reduce inter-subject variability. For biomarker discovery, all subjects were considered for the analyses.

2) Redundancy pruning: To reduce multicollinearity, pairwise absolute Spearman correlations $|\rho_{ij}|$ were computed and converted to a distance metric $d_{ij} = 1 - |\rho_{ij}|$ for Ward hierarchical clustering [18]. The resulting dendogram was cut at $d < 0.2$ (equivalent to $|\rho| > 0.8$) to group redundant features. From each cluster, a single representative feature (the medoid, i.e., the feature minimizing the mean intra-cluster distance) was retained.

3) Wasserstein-Based Modality-Invariance Ranking: To identify biomarkers robust to sensory shifts, the 1–Wasserstein distance (W_1) was used, which provides a robust geometric measure of divergence between distributions [20]. Let $X_{f,s}$ be the random variable for feature f and subject s. The modality set is defined as $\mathcal{M} = \{\text{text}, \text{audio}, \text{video}\}$, with unique pairs $\mathcal{C} \subset \mathcal{M}^2$.

A ranking criterion $\mathcal{J}(f)$ is defined to penalize modality-driven distribution shifts while favoring attention-state separability. First, the *inter-modality divergence* $\delta_{f,s}^{\text{shift}}$ is computed as the mean pairwise distance between modality-conditional distributions:

$$\delta_{f,s}^{\text{shift}} = \frac{1}{|\mathcal{C}|} \sum_{(u,v)\in\mathcal{C}} W_1\left(P(X_{f,s}|u), P(X_{f,s}|v)\right). \quad (1)$$

Second, *state discriminability* $\delta_{f,s}^{\text{sep}}$ between attention (Att) and relaxed (Rel) states is computed:

$$\delta_{f,s}^{\text{sep}} = W_1\left(P(X_{f,s} \mid \text{Att}), P(X_{f,s} \mid \text{Rel})\right). \quad (2)$$

Here, $P(X_{f,s} \mid \text{Att})$ and $P(X_{f,s} \mid \text{Rel})$ denote state-conditional distributions pooled across all sensory modalities, ensuring that state separability is evaluated independently of modality.

The final ranking score balances these factors, aggregated robustly across the training subject pool $\mathcal{S}$ via the median:

$$\mathcal{J}(f) = \frac{\underset{s\in\mathcal{S}}{\text{median}}\left(\delta_{f,s}^{\text{shift}}\right)}{\underset{s\in\mathcal{S}}{\text{median}}\left(\delta_{f,s}^{\text{sep}}\right) + \varepsilon}, \quad (3)$$

where $\varepsilon = 10^{-6}$ ensures numerical stability. Lower $\mathcal{J}(f)$ prioritizes features that minimize domain shift while maximizing class separability. During biomarker discovery, this ranking identifies a modality-stable "common core" of biomarkers for physiological interpretation.

4) Biomarker Discovery and Statistical Inference: To ensure statistical validity given the hierarchical structure of EEG data, inference was performed at the subject level rather than pooling epochs [12]. For a feature f and subject s, the subject-level contrast $\delta_{f,s}$ was defined as the difference in central tendency between states:

$$\delta_{f,s} = \tilde{x}_{f,s}^{\text{att}} - \tilde{x}_{f,s}^{\text{rel}}, \quad (4)$$

where $\tilde{x}_{f,s}^{(\cdot)}$ denotes the median of epoch-level samples for the respective condition, which is less sensitive to occasional large-amplitude artifact epochs that may remain in consumer-grade EEG.

a) Effect Size Quantification:: Given the modest cohort size, effect magnitude was quantified using Hedges' g, which applies a small-sample correction to reduce positive bias in small-N settings [54]:

$$g = d \times \left(1 - \frac{3}{4(N-1)-1}\right). \quad (5)$$

A strict effect size threshold of $|g| \geq 0.5$ (medium effect) was imposed for all candidate biomarkers.

b) Statistical Inference:: The condition effect was assessed using a hierarchical bootstrap ($B = 1000$ resamples) and validated with a Wilcoxon signed-rank test. To evaluate cross-modal consistency, modality-specific p-values were combined using Stouffer's Z-score method [22]. We chose Stouffer's method (instead of Fisher's combined probability test) because it combines the independent modality tests in a more balanced way, and it avoids letting one extremely small p-value from a single dominant modality drive the overall conclusion. Let $p_{f,k}$ be the two-sided p-value for feature f in modality k. The combined statistic Z_f was computed as follows:

$$Z_f = \frac{1}{\sqrt{K}} \sum_{k=1}^{K} \Phi^{-1}\left(1 - \frac{p_{f,k}}{2}\right), \quad (6)$$

where K is the number of modalities and $\Phi^{-1}(\cdot)$ is the inverse standard normal CDF (probit function). The final combined significance is given by $P_f = 2(1 - \Phi(|Z_f|))$.

Multiple comparisons were controlled using Benjamini–Hochberg FDR ($q = 0.05$) applied to the **median p-value** obtained across the LOSO resampling folds [52]. This robust aggregation prevents outlier folds from driving significance. Features were retained only if they survived FDR correction, met the effect size threshold ($|g| \geq 0.5$), and were significant in at least 75% of LOSO resampling iterations.

E. Evaluation Strategy: Explanatory vs. Predictive Modeling

Following the architecture in Fig. 1, the evaluation is bifurcated into two objectives:

- **Generalization stress test (LOSO–LOMO):** Assess the robustness of the discovered biomarkers under simultaneous subject and modality shift.
- **Leakage-controlled predictive validation (LOSO):** Validate the discovery methodology by enforcing fold-local selection and strict subject-level separation.

TABLE II
BIOMARKER DISCOVERY AND VALIDATION CRITERIA.

Stage	Specification
Normalization	Per-subject z-score (epoch-level)
Redundancy pruning	Spearman $\lvert\rho\rvert \leq 0.8$
	Ward clustering (medoid retained)
Feature Selection	1–Wasserstein ranking
	Effect size $\lvert g\rvert \geq 0.5$
Statistical inference	Hierarchical bootstrap ($B = 1000$)
	Wilcoxon signed-rank
Correction Factor	Hedges' g (small-sample bias corr.)
Multiple testing	Benjamini–Hochberg FDR ($q = 0.05$)
Robustness criterion	Significant in $\geq 75\%$ LOSO folds
Leakage control	All selection/inference within training folds

a) Interpreting balanced accuracy under strict real-world constraints: Balanced accuracies in the mid-0.60 range are expected under the present constraints: strict cross-subject LOSO evaluation (often 10%–20% lower than intra-subject schemes), consumer-grade 14-channel EEG with a higher noise floor, and single 6-second epoch decoding without time-averaging. In this setting, a statistically reliable lift from the 0.50 chance level to a median balanced accuracy of 0.66 reflects robust signal extraction and modality-tolerant biomarkers rather than a failure of predictive modeling.

1) Biomarker Stress Test: LOSO–LOMO Robustness: The purpose of this stress test is to evaluate the cross-modal and cross-subject stability of the specific biomarkers identified during the Phase 1 explanatory analysis.

As illustrated in the Phase 1 workflow, the final biomarker set is fixed *a priori* based on the global statistical inference (Stouffer's Z and false discovery rate (FDR)). These candidates are subjected to a LOSO–LOMO (Leave-One-Subject-Out, Leave-One-Modality-Out) protocol to determine whether they generalize to unseen subject–modality pairs (s^*, m^*). For each fold, training is performed on $N - 1$ subjects while strictly excluding the test modality m^*. This yields $14 \times 3 = 42$ independent evaluation folds, and the analysis evaluates whether these *explanatory* signals retain *predictive* power under extreme distribution shifts.

2) PHASE 2: Leakage-Controlled Predictive Modeling: This phase evaluates pipeline soundness under strict data leakage control. All modeling is performed using Leave-One-Subject-Out (LOSO) cross-validation, where each fold treats one subject as a fully held-out test set. Crucially, all preprocessing and feature selection steps are recomputed independently within each fold using only the training subjects ($N - 1$), with no access to the held-out subject.

To enforce a leakage-proof protocol [12], no population-level statistical inference from Phase 1 is reused in this phase. Hierarchical bootstrapping [53], Stouffer's combination, and false discovery rate (FDR) correction are therefore omitted; Phase 2 focuses on validating fold-local feature-selection stability rather than re-discovering global biomarkers.

Within each LOSO fold, feature selection on the training data consists of subject-wise z-score normalization, local redundancy pruning based on Spearman correlation, and fold-local 1-Wasserstein ranking to prioritize features that discriminate attentional state. Class imbalance is handled strictly by random downsampling of the majority class *on the training set only*.

3) Modality Classification: A negative-control task classifier was used to test whether the selected biomarkers reflect cognitive state rather than task/modality cues. Using leakage-controlled LOSO and *attention*-only epochs, Text/Audio/Video (task modalities) were classified with the fixed discovery-stage biomarker sets (no additional selection or tuning).

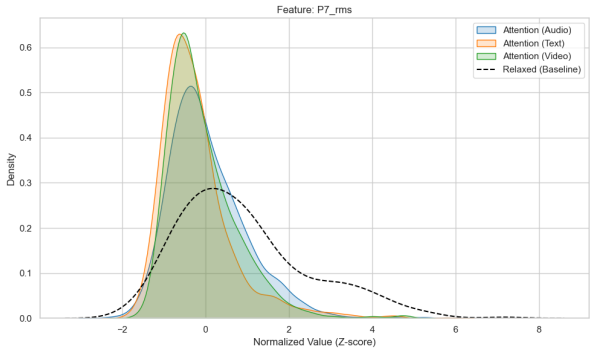


Fig. 2. Bootstrap distributions of global 1–Wasserstein distances (W_1) for *selected* candidate features. Each boxplot summarizes W_1 computed across bootstrap resamples for both attention–relaxed separation and inter-modality shift (text vs. audio vs. video). Selected features exhibit consistently non-zero state separation and comparatively small modality-driven shifts (lower median and tighter spread).

III. RESULTS

A. Feature Selection and Biomarker Discovery

Feature extraction yielded a pool of 611 features spanning time, frequency, complexity, energy, and spatial domains. Correlation-based redundancy pruning (threshold ≤ 0.20) reduced this pool to 322 non-redundant features while preserving discriminative information.

Features were then ranked by a 1–Wasserstein criterion balancing attention-vs-relaxed separability against modality-driven shifts, computed globally with 200 bootstrap resamples per feature. The top 83 features were retained for downstream validation (Figs. 2–3). Figs. 2 and 3 summarize these bootstrap distributions: *strong separation* corresponds to an attention-vs.-relaxed distance distribution that is consistently shifted away from zero (with a comparatively tight spread), whereas *modality sensitivity* manifests as large or highly variable distances between modality-specific distributions (e.g., text vs. audio vs. video). Selected features therefore show a reliable state-separation signal alongside comparatively small modality-driven shifts, while rejected features tend to show either weak state separation (distributions concentrated near zero) or pronounced modality-dependent variability.

B. Subject-Level Statistical Validation (LOSO)

The 83 redundancy-pruned and Wasserstein-ranked features were subjected to subject-level statistical validation in a

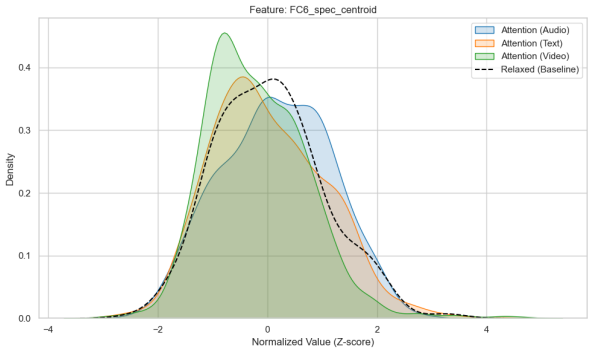


Fig. 3. Bootstrap distributions of global 1–Wasserstein distances (W_1) for *rejected* candidate features. Rejected features typically show weak attention–relaxed separability (distributions concentrated near zero) and/or large, variable inter-modality shifts, indicating modality sensitivity rather than a stable cross-modal attention biomarker.

TABLE III
FEATURE COUNTS ACROSS THE BIOMARKER DISCOVERY PIPELINE.

Stage	Number of Features
Initial feature extraction	611
After redundancy pruning	322
After Wasserstein ranking	83
After LOSO statistical validation (HBoot)	26
After LOSO statistical validation (Wilcoxon)	26
Final robust biomarkers (intersection)	26

leave-one-subject-out (LOSO) framework. Specifically, this work tested each candidate feature using two complementary nonparametric procedures—a hierarchical bootstrap ($B = 1000$ resamples per fold) [53] and a Wilcoxon signed-rank test applied to subject-level median differences—to reduce the chance that significance was driven by any single participant. Agreement between the bootstrap and Wilcoxon selections further increases confidence that the retained biomarkers are not artifacts of a particular inference procedure.

In both cases, false discovery rate (FDR) correction ($q \leq 0.05$) was applied to the fold-wise p-values (Benjamini–Hochberg, across features within each LOSO fold), a minimum effect-size threshold (Hedges' g; $|g| \geq 0.5$), and a survival criterion of 0.75, defined as the proportion of LOSO folds in which a feature remained FDR-significant.

C. Final Cross-Modal Robust Biomarkers

Both statistical approaches returned the same final biomarker set in this dataset, suggesting that the selection is relatively insensitive to the choice of statistical test. From the 83 candidates, **26 features** satisfied all selection criteria, with survival rates consistently exceeding the 75% threshold (Table IV). Here, *survival* denotes the number of LOSO folds in which a feature remained FDR-significant after applying the fold-local effect-size threshold.

The selected biomarkers are distributed across frontal (AF3/AF4, F3/F4, FC5), parietal (P7/P8), and occipital (O1/O2) sensors, with minor contributions from the temporal region (T7). Table IV lists the complete set of 26 robust biomarkers.

TABLE IV
COMPLETE LIST OF 26 FINAL SELECTED CROSS-MODAL ROBUST BIOMARKERS (LOSO).

Feature (ch.: descriptor)	Median q	Effect (g)
P8: absolute α power	9.3×10^{-9}	-0.772
P8: Petrosian fractal dimension	3.6×10^{-4}	$+0.920$
AF4: θ/α power ratio	4.3×10^{-4}	$+1.143$
O2: absolute α power	5.5×10^{-4}	-0.691
P7: Petrosian fractal dimension	5.5×10^{-4}	$+0.646$
P7: relative α power	5.5×10^{-4}	-0.666
O1: absolute α power	5.5×10^{-4}	-0.597
AF3: θ/α power ratio	5.5×10^{-4}	$+1.008$
P7: absolute θ power	5.5×10^{-4}	-0.544
F4: relative α power	9.4×10^{-4}	-0.599
P7: absolute α power	1.3×10^{-3}	-0.546
P7: θ/α power ratio	1.3×10^{-3}	$+0.985$
P8: relative α power	1.3×10^{-3}	-0.718
AF3: Petrosian fractal dimension	1.5×10^{-3}	$+0.721$
P7: RMS amplitude	1.5×10^{-3}	-0.696
F4: θ/α power ratio	1.5×10^{-3}	$+0.820$
AF3: relative α power	1.6×10^{-3}	-0.587
O2: Petrosian fractal dimension	2.0×10^{-3}	$+0.567$
FC5: relative α power	2.0×10^{-3}	-0.767
O1: relative δ power	2.1×10^{-3}	$+0.752$
F3: relative α power	2.5×10^{-3}	-0.767
O2: relative α power	2.6×10^{-3}	-0.692
T7: RMS amplitude	4.1×10^{-3}	-0.661
O1: relative α power	5.4×10^{-3}	-0.632
P7: relative δ power	7.1×10^{-3}	$+0.711$
T7: absolute α power	8.8×10^{-3}	-0.582

Note: Median q-values are reported following Benjamini–Hochberg FDR correction. Effect sizes are reported as Hedges' g, with sign indicating the direction of change (attention vs. relaxed).

The results highlight three dominant physiological patterns associated with the transition to a high-attention state:

- **Alpha suppression:** A widespread reduction in absolute and relative α power (e.g., P8, O2, P7, F3), consistently indicated by negative effect sizes ($g < -0.5$) [25], [26].
- **Increased complexity:** Higher Petrosian fractal dimension (e.g., P8, P7, AF3, O2) during attention states, suggesting increased neural information processing.
- **Spectral balance shift:** Increased θ/α power ratios (e.g., AF4, AF3, F4, P7), reflecting a shift from relaxed alpha dominance to focused theta activity.

This expanded set of 26 biomarkers confirms that the transition from a relaxed to an attentive state is characterized by a global desynchronization of alpha rhythms coupled with increased signal complexity, a finding that is robust across subjects and modalities [25], [26].

D. LOSO–LOMO Robustness and Classification Performance

To isolate the contribution of each stage in the discovery pipeline, this work evaluated five feature sets under nested LOSO–LOMO cross-validation ($n = 42$ subject–modality splits):

1) **Redundancy-pruned baseline:** Features surviving collinearity checks but without supervised selection.
2) **Wasserstein-ranked candidates:** Top-ranked features by modality divergence, before statistical filtering.
3) **FDR-validated set:** Features passing Benjamini–Hochberg correction ($q < 0.05$).

TABLE V

STEPWISE EVALUATION OF BALANCED ACCURACY (MEAN ± STANDARD DEVIATION) ACROSS LOSO–LOMO FOLDS ($n = 42$). THE FINAL SET ATTAINS THE BEST SVM PERFORMANCE BY COMBINING STATISTICAL SIGNIFICANCE, EFFECT MAGNITUDE, AND CROSS-SUBJECT STABILITY.

Feature Set	SVM (RBF)	LDA
Redundancy-pruned	0.597 ± 0.094	0.604 ± 0.137
Wasserstein selected	0.598 ± 0.135	0.619 ± 0.110
FDR-validated ($q < 0.05$)	0.620 ± 0.140	0.611 ± 0.089
FDR + effect size ($\|g\| > 0.5$)	0.620 ± 0.155	0.605 ± 0.121
Final biomarkers (survival > 0.75)	$\mathbf{0.641 \pm 0.133}$	0.601 ± 0.116

4) **FDR & effect-validated set:** Features with $q < 0.05$ and a medium effect-size threshold ($|g| > 0.5$).
5) **Final biomarker set:** Intersection of $q < 0.05$, $|g| > 0.5$, and the stability criterion (survival rate > 0.75).

This work reports results for an SVM (RBF kernel) and LDA to contrast non-linear versus linear separability.

1) Impact of Stepwise Filtering: Table V summarizes performance across feature sets. The **Wasserstein-ranked** candidates yielded an SVM balanced accuracy of 0.598 ± 0.135. Applying **FDR correction** removed statistical noise and increased SVM performance to 0.620 ± 0.140.

Interestingly, adding the **effect-size** constraint ($|g| > 0.5$) without stability filtering did not further improve mean SVM performance (still 0.620) and increased the standard deviation to 0.155.

The main improvement came from the **stability (survival)** criterion. The **final robust biomarker set** achieved the best SVM performance ($\mathbf{0.641 \pm 0.133}$) while reducing variance relative to the effect-size-only set. This indicates that the survival threshold removes volatile, large-effect features and retains those that generalize across subjects.

In contrast, LDA performance peaked with the Wasserstein set (0.619) and declined as filters became stricter (0.601 for the final set). This divergence suggests that the pipeline distilled non-linear interactions captured by the SVM, whereas the linear trends captured by LDA were more likely driven by noise or modality-specific artifacts.

Figure 4 visualizes these distributions. The intermediate sets (FDR and effect size) improve over the redundancy-pruned baseline, but the final biomarker set offers the best trade-off: higher median balanced accuracy and reduced lower-tail failures relative to the effect-size-only filter.

E. Leave-One-Subject-Out (LOSO) Classification Validation

To evaluate subject-independent generalization, strict leave-one-subject-out (LOSO) classification was performed. In each fold, all epochs from one subject were held out for testing, while models were trained exclusively on the remaining subjects. Support Vector Machines (SVM) and Linear Discriminant Analysis (LDA) were treated as primary classifiers, representing a strong non-linear baseline (RBF SVM) and a simpler linear baseline (LDA), consistent with standard model-selection practice in small-sample classification citehastie2009elements.

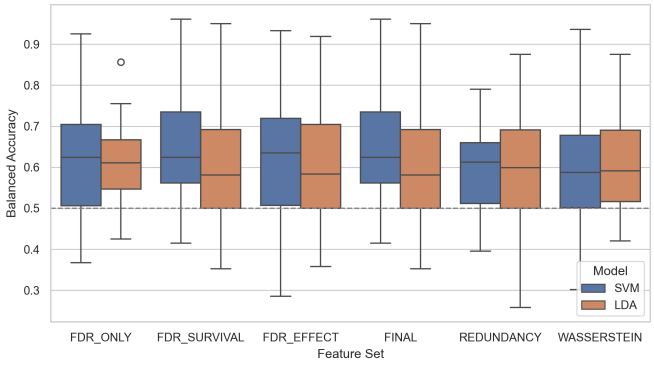


Fig. 4. Distribution of balanced accuracy across five discovery stages under nested LOSO–LOMO validation ($n = 42$). The final biomarker set (survival > 0.75) attains the best SVM performance (0.641 ± 0.133) and improved cross-modal stability relative to intermediate statistical and effect-size filters.

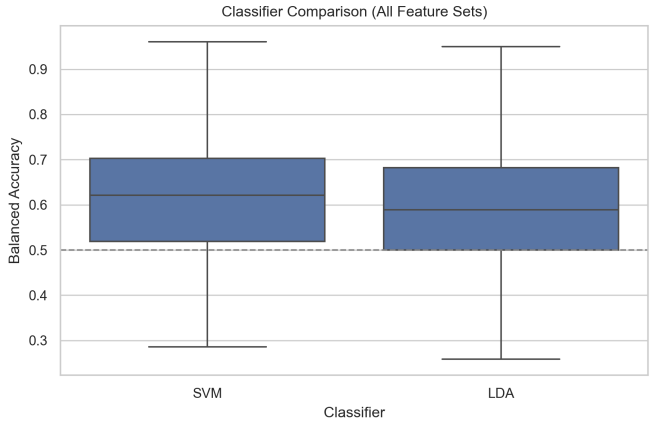


Fig. 5. Comparative performance of SVM and LDA classifiers. The SVM benefits most from stability filtering, whereas LDA changes little, consistent with non-linear structure in the identified biomarkers. The dashed red line indicates chance (0.5).

All feature processing was performed fold-locally to prevent information leakage, including subject-wise robust z-score normalization, redundancy pruning, and Wasserstein-based feature ranking, each fit using training data only. Three feature sets were evaluated: redundancy-pruned features, Wasserstein-selected features, and the final robust biomarker set. Because attention and relaxed classes were imbalanced, performance was quantified using median balanced accuracy across LOSO folds (chance level $= 0.5$), which provides a robust estimate under inter-subject variability.

Table VI reports median LOSO balanced accuracy across folds. The final robust biomarker set achieves the strongest and most consistent performance, with median balanced accuracy of 0.66 for SVM and 0.6 for LDA.

1) Modality Classification Results: In the task classification, LOSO SVM task classification on *attention*-only epochs was only modestly above chance (balanced accuracy 0.43–0.50

TABLE VI
MEDIAN LOSO BALANCED ACCURACY ACROSS FEATURE SETS.

Feature Set	LDA	SVM
Redundancy-pruned	0.52	0.63
Wasserstein-selected	0.61	0.64
Final robust biomarkers	0.60	**0.66**

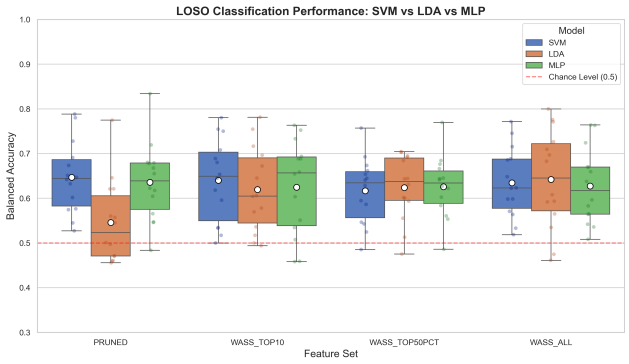


Fig. 6. Balanced-accuracy distributions across leave-one-subject-out (LOSO) folds for SVM and LDA under different feature-selection strategies (redundancy pruning vs. Wasserstein-based top-10, top-50%, and all selected features). Boxplots summarize subject-wise balanced accuracy (median and spread), with points marking individual held-out subjects; the dashed red line indicates chance (0.5). Wasserstein-based selection yields higher and more stable performance, with SVM achieving the best median accuracy.

across feature sets; 0.44 for the final biomarkers vs. chance 0.33), indicating weak task/modality discrimination and largely task-independent attentional encoding.

IV. DISCUSSION

A cross-modal biomarker-discovery pipeline for sustained attention in text, audio, and video learning was developed. Subject-wise normalization, redundancy pruning, and Wasserstein-based modality-invariance ranking yield a compact feature set that remains discriminative under modality shift. In the reported experiments, the final robust biomarker set achieved a median LOSO balanced accuracy of 0.66 with SVM (Table VI) and showed above-chance performance under the stricter LOSO–LOMO robustness test (SVM: 0.64; Table V). The selected biomarkers—dominated by frontal/parietal θ/α ratios and Petrosian fractal dimension—are consistent with prior EEG attention-decoding work linking spectral balance and complexity-related features to cognitive engagement, while the subject-wise spread in Fig. 6 highlights the heterogeneity typical of consumer-grade recordings.

A. Limitations

Limitations include the modest cohort size ($N = 14$), consumer-grade EEG signal quality and artifact susceptibility [42], a narrow participant demographic (young healthy adults), and labels derived from task instructions rather than continuous ground truth.

These issues are partially mitigated through artifact-aware preprocessing (ICA/ICLabel and AutoReject), subject-level inference, strict fold-local fitting/selection, and explicit robustness tests (LOSO and LOSO–LOMO with the final biomarker set fixed *a priori*).

B. Practical implications and next steps

The proposed pipeline supports reproducible, interpretable biomarker discovery under realistic acquisition constraints. With further validation, compact feature sets (Table IV) could enable online attention monitoring in adaptive learning or human–computer interaction. Because labels currently contrast attentive engagement with a relaxed baseline, a key next step is to move beyond binary classification toward graded or continuous attention measures (e.g., moment-to-moment engagement estimated from behavior or physiology).

Future work will emphasize replication in larger and more diverse cohorts, improved supervision and baseline control, and acquisition protocols that reduce artifact-driven variability, alongside hybrid or personalized models to better accommodate inter-subject differences.

V. CONCLUSION

This paper presents a framework for discovering cross-modal EEG biomarkers of sustained attention using consumer-grade hardware. By combining redundancy pruning, Wasserstein-based modality-invariance ranking, and strict subject-wise validation, the proposed approach produces a concise and interpretable feature set that generalizes to held-out participants in this dataset (median LOSO balanced accuracy: 0.66 with SVM) and remains above chance under a LOSO–LOMO stress test (SVM: 0.64). Overall, these results highlight the importance of fold-local fitting/selection in small-N EEG studies and motivate further validation in larger, more diverse cohorts and in settings with richer (and potentially continuous) attention labels.

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